Log-Scale Class Weighting for Robust Imbalanced Classification

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Abstract

Class imbalance is a pervasive challenge in machine learning, arising across diverse domains such as medical diagnosis, fraud detection, and visual recognition, where minority classes are critically important yet severely underrepresented. Weighted Cross-Entropy (WCE), the most direct remedy, assigns inverse-frequency weights to each class, but these weights diverge rapidly as the imbalance ratio grows, leading to numerical instability. More sophisticated alternatives within the re-weighting family, such as Class-Balanced (CB) Loss and Focal Loss, introduce additional hyperparameters and can degrade below standard cross-entropy (CE) performance at high imbalance ratios. We propose **Logarithmic Weighted Cross-Entropy (LWCE)**, which replaces inverse-frequency weights with logarithmically compressed counterparts $w_i = 1/\log(n_i + 1)$, and its generalization **PLWCE** ($w_i = 1/(\log(n_i + 1))^{\alpha}$), which allows dataset-adaptive tuning of the focusing strength via a single exponent α selected by Optuna grid search on a small proxy subset. We conduct a controlled evaluation on CIFAR-10-LT and CIFAR-100-LT at imbalance ratios of 10, 50, and 100, using ResNet-32 with five random seeds. Comparing six loss functions (CE, PWCE, LWCE, PLWCE, CB Loss, Focal Loss), we find that LWCE and PLWCE consistently outperform CE on macro-averaged F1 and tail-class accuracy as the imbalance ratio increases, while CB Loss and Focal Loss degrade relative to CE at IR = 100. On CIFAR-10-LT at IR = 100, PLWCE achieves F1-Macro of 0.458±0.018 versus 0.431±0.021 for CE and 0.406±0.015 for CB Loss. On CIFAR-100-LT at IR = 100, LWCE and PLWCE reach F1-Macro of 0.185 and 0.186, surpassing CE (0.179) and CB Loss (0.158). These results suggest that logarithmic weight compression offers a simple, stable, and parameter-efficient alternative to existing re-weighting strategies for class-imbalanced classification.

1 Introduction

In many real-world classification tasks—spanning medical diagnosis, financial fraud detection, network intrusion detection, and visual recognition—sample counts across categories follow a long-tailed distribution: a small number of majority classes concentrate the bulk of training data while many minority classes are represented by only a handful of examples [5, 3]. Under such conditions, a classifier trained with standard cross-entropy (CE) loss is implicitly biased toward majority classes: the aggregate gradient signal is dominated by numerically abundant samples, so the model achieves high overall accuracy by largely ignoring rare but equally valid categories—a phenomenon commonly termed *majority bias.*

Existing strategies to combat class imbalance fall broadly into two families. **Resampling methods** directly manipulate the training distribution: oversampling techniques such as SMOTE [4] synthesize new minority-class instances, while undersampling discards majority-class examples to restore balance. Although conceptually straightforward, both directions carry inherent risks. Oversampling can introduce synthetic artifacts and lead to overfitting on interpolated points, whereas undersampling discards potentially informative majority-class samples, distorting the true data distribution. Algorithm-level approaches avoid data manipulation altogether by modifying the training objective to reflect class importance. These include margin-based methods that widen decision boundaries for minority classes [3], logit-adjustment techniques that calibrate class-prior probabilities at inference time, and *loss re-weighting* methods that scale the cross-entropy loss by class-dependent weights. While the former two families require architectural modifications or post-hoc calibration, re-weighting methods are architecture-agnostic and applicable across data modalities with minimal implementation overhead. This paper focuses on the re-weighting family.

Weighted Cross-Entropy (WCE) [18] scales the loss for class i by $w_i \propto 1/n_i$, where n_i is the number of training samples in that class. Although conceptually simple, WCE suffers from a critical numerical instability: as the imbalance ratio (IR) grows, the inverse-frequency weight $1/n_i$ diverges rapidly, producing extreme gradient magnitudes that destabilize optimization. This becomes especially problematic at severe imbalance—for instance, in CIFAR-100-LT at IR = 100 the rarest class contains only $n_i = 5$ samples, making straightforward WCE impractical.

Several more sophisticated alternatives have been proposed. **Focal Loss** [13] addresses the class imbalance problem at the *sample* level by introducing a modulating factor $(1-p_t)^\gamma$ that down-weights easy, often majority-class examples. However, its behavior depends on per-sample prediction probabilities rather than class frequency, making it sensitive to the calibration quality of the model and yielding inconsistent gains across different imbalance regimes. **Class-Balanced (CB) Loss** [5] grounds re-weighting in the concept of *effective number of samples*, providing a theoretically motivated schedule through a hyperparameter $\beta \in (0, 1)$. Yet β requires careful tuning, and in our experiments CB Loss consistently underperforms CE at high imbalance ratios—degrading to F1-Macro of 0.406 against CE's 0.431 on CIFAR-10-LT at IR = 100, and 0.158 against 0.179 on CIFAR-100-LT at the same ratio—suggesting that its weight schedule becomes too aggressive when tail-class sample counts are very small.

To overcome the weight explosion of WCE while retaining its simplicity and interpretability, we propose **Logarithmic Weighted Cross-Entropy (LWCE)**, which replaces the inverse-frequency weight with a logarithmically compressed counterpart:

$$w_i = \frac{1}{\log(n_i + 1)}. \tag{1}$$

The logarithmic compression fundamentally bounds weight growth: whereas WCE weights grow as $\mathcal{O}(n_i^{-1})$, LWCE weights grow only as $\mathcal{O}((\log n_i)^{-1})$. The method preserves a meaningful boost for rare classes without inducing numerical instability, and requires no additional hyperparameters. We further generalize LWCE to **Power-Log Weighted CE (PLWCE)** by introducing a single tunable exponent α:

$$w_i = \frac{1}{\big(\log(n_i + 1)\big)^{\alpha}}, \tag{2}$$

which spans a continuous spectrum from near-uniform weighting ($\alpha \to 0$) to aggressive minority focus ($\alpha \gg 1$), with LWCE recovered at $\alpha = 1$. The optimal α per imbalance ratio is identified via Optuna grid search on a 20% proxy subset, requiring negligible additional computation.

We evaluate the proposed methods on the standard long-tailed image classification benchmarks, CIFAR-10-LT and CIFAR-100-LT, across three imbalance ratios (IR $\in \{10, 50, 100\}$) using ResNet-32 trained from scratch [8]. All experiments are repeated with five random seeds to report mean $\pm$ standard deviation. We compare LWCE and PLWCE against CE, Power-Scaled WCE (PWCE), CB Loss, and Focal Loss using overall Top-1 accuracy, macro-averaged F1-score, and group-wise accuracy on head (Many), middle (Medium), and tail (Few) classes defined by a tertile split of training counts.

Results demonstrate that LWCE and PLWCE consistently improve over CE on F1-Macro and Few-class accuracy as the imbalance ratio increases, while avoiding the performance degradation exhibited by CB Loss and Focal Loss at high IRs. On CIFAR-10-LT at IR = 100, PLWCE achieves F1-Macro of 0.458 ± 0.018, surpassing CE (0.431 ± 0.021) and CB Loss (0.406 ± 0.015). On CIFAR-100-LT at IR = 100, LWCE and PLWCE reach F1-Macro of 0.185 and 0.186, outperforming CE (0.179) and CB Loss (0.158), with the gap widening at the tail-class level.

The main contributions of this paper are:

1. We formally characterize the *weight explosion* failure mode of WCE and demonstrate its practical consequences on CIFAR-LT benchmarks.

2. We propose LWCE, a parameter-free drop-in replacement for WCE based on logarithmic weight compression, and its single-parameter generalization PLWCE.

3. We provide a controlled comparison of six loss functions on CIFAR-10-LT and CIFAR-100-LT with five-seed repetition, revealing that logarithmic re-weighting offers a more stable and consistent improvement over CE than CB Loss or Focal Loss, particularly in the high-imbalance regime.

The remainder of this paper is organized as follows. Section 2 reviews related work on long-tailed learning. Section 3 formally defines LWCE and PLWCE and analyzes their theoretical properties. Section 4 describes the experimental setup and presents results. Section 5 concludes.

2 Related Work

3 Proposed Method

4 Experiments

5 Conclusion

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